PP-HFM: Patient-Private Hierarchical Federated Multimodal Learning

Project report: the idea, the system, every experiment and what each one showed

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This is an internal report, not the paper. Every number is taken from the result files under `SIMULATORS/results/` (index: `RESULTS_FINAL.md`). All planned experiments are complete.

1 The project in one page

The one-sentence version. We built a three-tier federated learning system (devices → edge servers → cloud) that trains a multimodal clinical model without moving raw data, and that protects every *patient*, not just every data *record*, with a mathematically proven differential-privacy guarantee. We then measured, carefully, what that protection costs.

The problem. Hospitals and IoT devices produce data that cannot leave the site. Federated learning (FL) solves this: each site trains locally and only sends model updates. Differential privacy (DP) adds noise so that the updates do not reveal any individual. But almost every published DP system protects a *record* (one 10-minute window of sensor data). A patient contributes many records, around 8 in VitalDB and 28 in eICU. Protecting each record does *not* protect the patient: if you convert after the fact ("group privacy"), a guarantee of $\varepsilon \approx 1$ per record becomes $\varepsilon \approx 8$ per patient, which is almost no protection.

Our answer. PP-HFM makes the patient the privacy unit from the start. All of a patient's records are summed into one gradient, that sum is clipped, and noise is added to it. Each client (hospital) gets its own noise level calibrated to its own patient count. Every release in the hierarchy (devices, edges, cloud) is accounted for, and a theorem proves the guarantee.

What we found, in brief.

1. **The system works.** Uplink traffic falls 90.8–96.1 % (measured payload), energy per round falls 72.3–79.6 % (CloudSim Plus model), and inference takes at most 2.3 ms on a workstation CPU (target was 100 ms), on all three datasets.
2. **Patient-level privacy at $\varepsilon = 0.9$ costs no accuracy on VitalDB**, split so that each person is in one split and on one client (AUROC 0.832 private vs 0.820 non-private). On eICU it costs 3.1 AUROC points, which is inconclusive against the 2-point target, and 3.9 points end to end, which misses it. The cost comes from the smallest hospitals.
3. **The thing that makes strict privacy survivable is pretrained initialization**, not the trick of privatizing only the model head. A randomly initialized private model loses 0.19 AUROC; a pretrained one loses 0.004.
4. **The cost of protecting patients grows with the number of records per patient, k, and then flattens.** This follows the curve $A(1 - 1/k)$, and vanishes at $k = 1$, where a patient *is* a record. It was forecast in advance and tested on a second hospital corpus: the *shape* transferred, the *size* did not.
5. **The privacy audit separates the two units.** On VitalDB, an attack with a working positive control finds record-level training above its difficulty floor (LiRA AUC 0.576) and patient-level training at it (0.503).
6. **Against two re-implemented prior systems** (HED-FL, FL2DP), PP-HFM uploads 2.2–4.0 times less and shows lower measured leakage in the VitalDB audit. On eICU, non-private HED-FL is more accurate (0.825 vs 0.794).
7. Several things we expected did *not* hold, and we report them: the attack difference is smaller than predicted (0.58, not 0.7); private conformal prediction gave useless prediction sets; fusion did not help on eICU's natural streams.

2 How the project progressed (phases)

Phase	What happened
1–2	Literature review and problem definition. Four research gaps identified (Section 2.1); four quantified objectives fixed.
3	Environment: Python stack, NS-3 network simulator (v3.46.1), CloudSim Plus (v8.5.6) for energy, GPU utilities.
4	Both simulators implemented: network (latency, traffic, reliability) and edge–cloud resource/energy model. First system-level numbers.
5	The learning system: multimodal model, hierarchical FL, DP, compression, conformal calibration. All 17 deliverables on VitalDB and WEAR (5 Aug). Four silent bugs found and fixed.
6	Ablations and baselines: which component does what. Hierarchy resolved at 6 seeds; pretrained-initialization finding.
7	External validation on eICU (187 hospitals), the k-scaling law, pre-registered tests, paper writing.
Fix (22–29 Oct)	A flaw in the patient-DP mechanism was found while writing the proof. Fixed, and every patient-level experiment was re-run in stages (Section 6).
Baselines (29–30 Oct)	HED-FL and FL2DP re-implemented and run under the same data, encoders and seeds.
Person-level (30 Oct–1 Nov)	VitalDB had been split by surgical case, so a person could be in two splits. Rebuilt the split by person and re-ran the privacy cost, the cost of the unit, the baselines and the audit.

2.1 The four research gaps

1. **No multimodal fusion in private FL**: existing systems treat data as generic gradients.
2. **Privacy vs. efficiency unresolved**: HED-FL has a hierarchy but no DP; FL2DP has DP but no hierarchy.
3. **Real-time QoS** (< 100 ms) is not addressed in robust FL frameworks.
4. **The privacy unit**: guarantees are per record, while clinical deployments must protect people.

2.2 The four objectives

Objective	Target	Measured by
O1 Uplink traffic reduction	$\geq 50\,\%$	measured payload of the update encoding
O2a Privacy budget	$\varepsilon \leq 1$	privacy accountant
O2b Utility loss from privacy	≤ 2 points	paired training runs
O3a Inference latency (P95)	< 100 ms	ONNX Runtime, workstation CPU
O3b Update delivery	$> 99\,\%$	ns-3 simulation
O4 Energy saving per round	$\geq 20\,\%$	CloudSim Plus model

Each objective is measured by exactly one tool, so no number is reported twice from two sources.

3 The system

3.1 The three tiers

Tier 1 – clients (devices / hospitals). Each holds its own patients. It trains the model locally with patient-level DP-SGD (below) and sends a compressed update: only the top 20 % of values, quantized to 8 bits, with error feedback so nothing is permanently lost.

Tier 2 – edge servers. Each edge averages the updates of its clients, $\tau_e = 5$ times per cloud round. Most traffic therefore stays local and never crosses the wide-area link.

Tier 3 – cloud. Averages the edges once per round ($R = 10$ rounds) and broadcasts the global model.

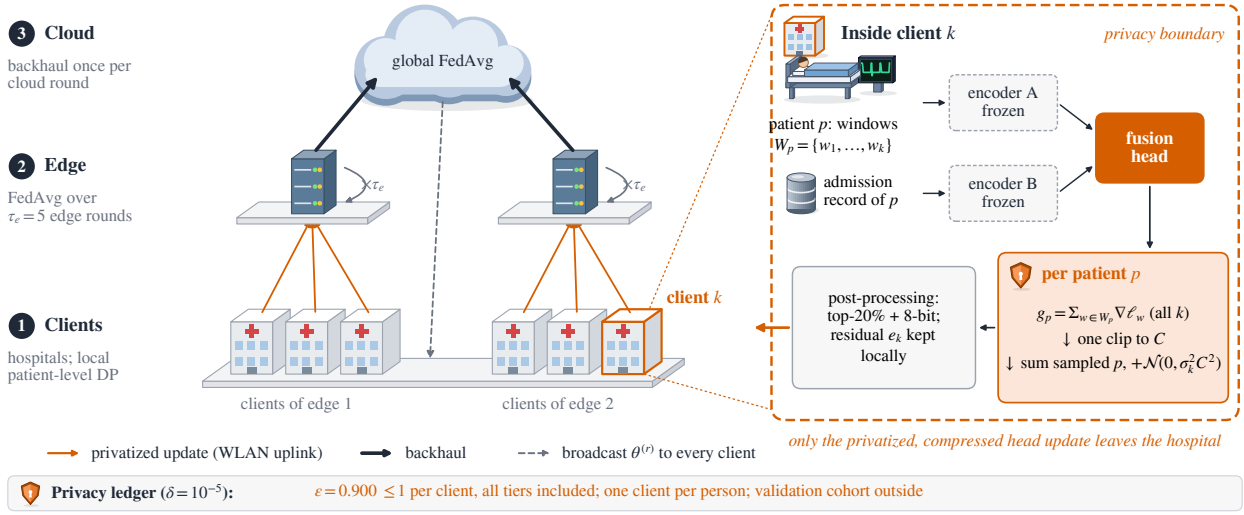


Figure 1: PP-HFM: clients train with patient-level DP, edges aggregate several rounds locally, the cloud aggregates the edges.

3.2 The model

Two encoders, one per modality (e.g. vital-sign time series and pre-operative clinical variables), whose embeddings are concatenated and passed to a small *fusion head*. The model has 1.86 M parameters; the head is 0.56 M (30 %). The encoders are pretrained once and frozen; federated training updates only the head. All normalization layers are group/layer norm, because batch norm mixes patients inside a batch and would break the per-patient privacy accounting.

3.3 Patient-level DP-SGD (the core mechanism)

1. **Sample patients, not records.** Each patient is included in a step independently with probability q (Poisson sampling).
2. **Sum per patient, then clip.** All of a sampled patient's record gradients are summed into one vector, and that vector is clipped to norm $C = 1$. One patient can therefore move the update by at most C, however many records they have.
3. **Add Gaussian noise** of scale $\sigma_k C$ and divide by the *expected* batch size $q|P_k|$ (not the realized one; see Section 6).
4. **Per-client noise.** Every patient belongs to exactly one client, so each client's noise σ_k can be calibrated to its own population (parallel composition), instead of forcing the worst hospital's noise on everyone. On eICU this means a median σ of 13.4 instead of 29.3 everywhere.
5. **Accounting.** The PRV accountant computes the total $(\varepsilon, \delta) = (0.9, 10^{-5})$ over all rounds and tiers.

Theorem 1 (what is proven). For declared public cohort sizes and fixed preprocessing, the full training transcript of every tier is (ε, δ)-DP with respect to adding or removing one patient. It covers the *training* process. It is end-to-end only if the encoders and the standardization constants are also fitted on public data; among our runs, that is the eICU re-run with pool-fitted constants (Section 5.16).

4 The data

	VitalDB	WEAR	eICU
Domain	surgery	wearable activity	intensive care
Clients	9 surgical specialties	22 participants	187 hospitals
Individuals	5,974 persons	22	132,736
Records per individual ($\bar{k}$)	7.9	$\approx$ 1,750	28.5
Modality A	9 vital trends + masks	inertial sensors	6 vitals + masks
Modality B	34 pre-op variables	2048-d video features	5 admission values
Task	ICU admission	18 activities	mortality
Positive rate	18.84 %	–	4.22 %

Two VitalDB splits. The *person-level* split puts each person in one split and on one client: 4,221 training, 868 validation and 885 test persons, after dropping 200 cases that would have put a person in two places. The earlier *case-level* split was by surgical case (6,257 cases, 19.65 % positive), so 55 of 940 test persons also had a training case. The privacy cost, the cost of the unit, the baselines and the audit use the person-level split; the other VitalDB analyses are case-level and say so.

Three choices protect the results from silent inflation: splits are by individual (never by window); only pre-operative variables are used (the intra-operative ones encode the outcome); missing values get an explicit indicator channel (an arterial line is placed only in patients judged at risk, so "missing" is itself informative). On eICU a pool of 526,867 windows from individuals who are in no client is reserved for pretraining the encoders, so eICU encoders never see protected patients. On VitalDB the cohort is too small for that, so the headline uses "oracle" encoders pretrained on client data, which is an upper bound (the cost of doing it properly is measured in Section 5.17).

5 Experiments, one by one

Each experiment below has the same four parts: the question, what we did, the result, and what it means. "Points" means percentage points of AUROC (0.01 AUROC = 1 point). Intervals are 95 % confidence intervals across seeds.

5.1 System objectives on three corpora

Question. Does the full system meet the four objectives?

What we did. Ran the complete system on VitalDB, WEAR and eICU; uplink measured from the encoded payload, network simulated in ns-3 at packet level on every round, energy modeled in CloudSim Plus, latency timed with ONNX Runtime on a workstation CPU (1,000 runs after 100 warm-up). Verdicts follow the rule fixed in advance: the confidence interval decides, not the point estimate.

Result.

	VitalDB	WEAR	eICU	Target
O1 Uplink reduction	90.8 %	96.1 %	92.3 %	≥ 50 %
O1b Round-latency cut	16.9 %	21.7 %	33.4 %	> 0
O2a Training ε (total)	0.900 (1.0)	0.766 (0.966)	0.900 (1.0)	≤ 1
O2b Utility loss	-1.18 AUROC	1.97 pp acc. (inconcl.)	3.10 AUROC (inconcl.)	≤ 2
O3a P95 latency	2.30 ms	0.75 ms	0.24 ms	< 100 ms
O3b Update delivery	100.00 %	99.94 %	97.83 % (**fails**)	> 99 %
Passed	7 of 7	6 of 7	5 of 7	
O4 Energy saving	72.3 %	76.8 %	79.6 %	≥ 20 %

What it means. Communication, latency and energy targets are met everywhere by a wide margin. VitalDB (person-level) passes everything. WEAR's utility interval spans the threshold (and is a record-level accuracy loss, not an AUROC cost). On eICU the utility interval also spans the threshold, with a point estimate above it, and delivery fails at two edges (187 clients sharing two edges, Section 5.3). "Total" adds the conformal

release, which was evaluated but is not part of the deployed system. The scores are in `objectives_*.json`; an earlier scoring is kept as `*_superseded.json`.

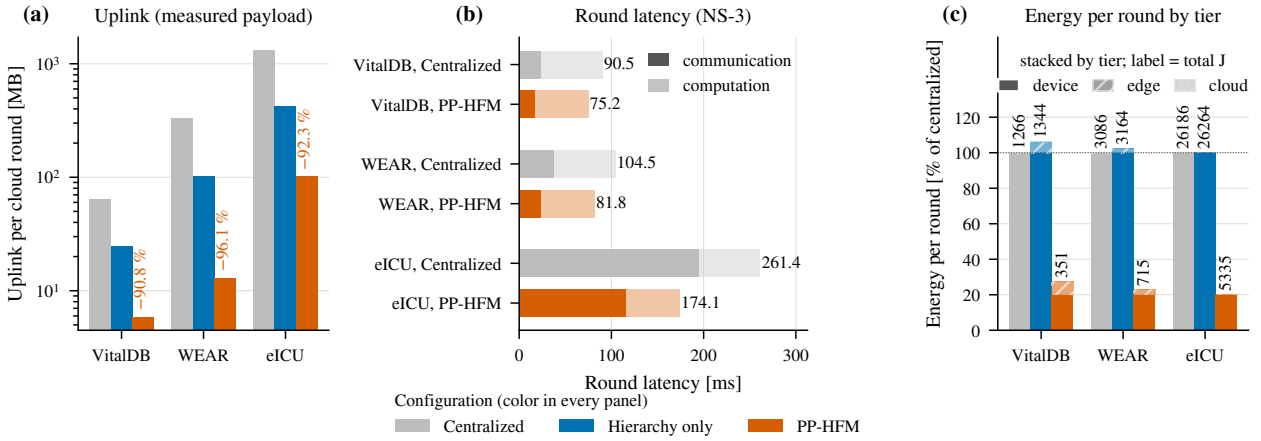


Figure 2: System cost relative to centralized training: uplink, round latency, energy by tier.

5.2 Inference latency

Question. Can the model answer within 100 ms at the edge?
What we did. Exported to ONNX, timed on a workstation CPU (Intel Core i7-14700), not on edge hardware; also tried 8-bit quantization.
Result. P95 latency 2.30 / 0.75 / 0.24 ms. On VitalDB 94 % of it is the forward pass. 8-bit quantization made it *slower* (P95 17.3 ms) and broke the WEAR GRU, so it was rejected.
What it means. More than a fortyfold margin. Latency is not a constraint for this model.

5.3 Update delivery vs. number of edges (eICU)

Question. Why does eICU miss the 99 % delivery target?
What we did. Kept everything fixed and varied the number of edge servers.
Result. With 2 edges, 187 clients contend for the wireless medium: 97.83 % (centralized: 95.11 %). Delivery rises monotonically, meets 99 % on average from 6 edges and in every round from 8 (about 23 clients per edge), reaching 99.94 % at 16.
What it means. It is a provisioning question, not a design flaw: the hierarchy needs enough edges for the client count.

5.4 Utility cost of privacy (O2b)

Question. How much accuracy does $\varepsilon = 0.9$ patient-level privacy cost?
What we did. Identical system with and without DP, 5 paired seeds, per-client noise, PRV accountant. AUROC is primary, because at the default 0.5 threshold the private model predicts the majority class on every seed (accuracy = base rate), so accuracy measures the threshold, not the model.
Result.

- **VitalDB (person-level):** AUROC 0.8318 private vs 0.8200 non-private, a change of -1.18 points $[-1.63, -0.73]$ ($p = 0.002$); AUPRC -3.47 $[-6.33, -0.61]$; balanced accuracy -0.31 $[-1.23, +0.62]$. Objective met. The private model is slightly *better*; this is consistent with clipping and noise regularizing the small head, but this experiment does not isolate why, so it is not read as "privacy helps". On the old case-level split the change was -0.90 $[-1.82, +0.02]$.
- **eICU** (one hospital per person): 0.8239 → 0.7929, a cost of 3.10 points $[1.55, 4.65]$. The interval contains 2, so by the rule fixed in advance the objective is *inconclusive*, although the point estimate is above the target. AUPRC cost 1.84 (interval spans 2), balanced-accuracy cost 0.33 (met). End to end (pool-fitted constants): 3.88 $[2.60, 5.17]$, a clear miss in AUROC and in balanced accuracy (2.80

[2.52, 3.08]); the AUPRC interval spans the target (Section 5.16).

- **WEAR** (record-level): accuracy loss 1.97 pp [1.76, 2.17]; the 2-point threshold lies inside the interval, so it is inconclusive. Seeds were fixed in advance at 15 and not extended.

What it means. On eICU the cost comes from the smallest hospitals: with a dozen patients they must sample every patient at every step and carry the largest noise multipliers (up to 29.3).

5.5 Cost of the privacy unit: patient vs. record

Question. At the same budget, what does it cost to protect patients instead of records?
What we did. VitalDB (person-level), $\varepsilon = 1$, 5 paired seeds, record-level DP vs patient-level DP.
Result. AUROC 0.8423 → 0.8254, a change of −0.0170 [−0.0269, −0.0071]; AUPRC −0.0582 [−0.0701, −0.0462]. (Case-level split: −0.0147 and −0.0783.) For comparison, converting a record-level guarantee to patients by group privacy would turn $\varepsilon = 1$ into $\varepsilon \approx 8$; direct patient-level calibration instead needs only 1.77× more noise.
What it means. Protecting the right unit has a real but moderate cost. Two causes: fewer units per step (4,221 patients vs 33,491 records), and clipping hurts patients with many records most. The k-law experiments isolate the second.

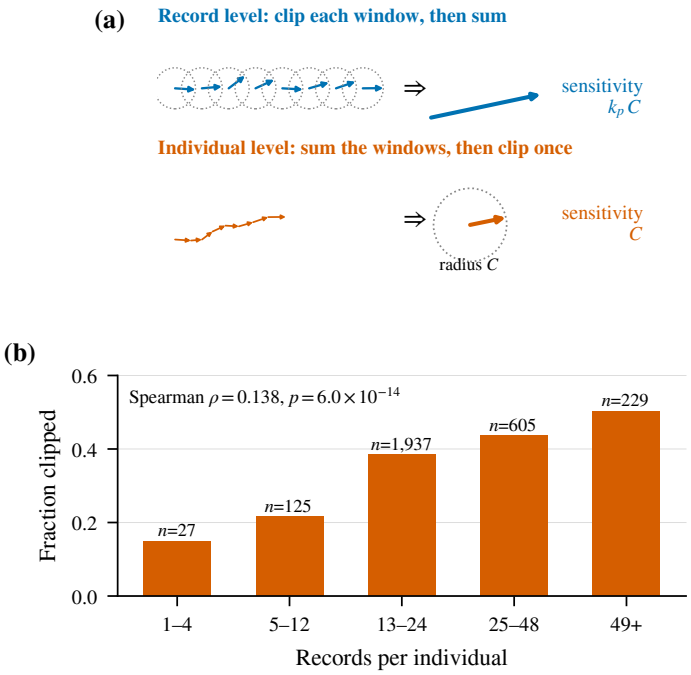


Figure 3: Record-level vs patient-level privacy unit.

5.6 What makes strict privacy survivable: initialization

Question. Our recipe privatizes only the head over pretrained frozen encoders. Is the benefit from "head-only" DP or from pretraining?
What we did. Three arms, so the two effects can be separated, at six budgets from $\varepsilon = \infty$ to 0.5: **B4** full model, random initialization (the usual prior art); **B7** full model, pretrained initialization; **B6** head only, pretrained (ours).
Result. (VitalDB case-level split and WEAR, patient-level mechanism on VitalDB, 3 seeds, after the fix.)

- B4 → B7 (pretraining): +0.013 at $\varepsilon = \infty$ rising to +0.199 at $\varepsilon = 0.5$; it grows at every one of the 5 steps.
- B7 → B6 (head-only): −0.003 to −0.006, every interval covers zero.
- From $\varepsilon = \infty$ to 0.5 the random model loses 0.19 AUROC, the pretrained one 0.004.

What it means. Pretraining does the work; head-only DP adds nothing to accuracy. The effect grows exactly as privacy tightens and nearly vanishes without privacy, a built-in dose–response. The value of head-only training is *communication*: only the head is transmitted, which is where the 90 % uplink saving comes from. (Note: freezing does not reduce the noise; noise depends on budget, steps and sampling, never on parameter count.)

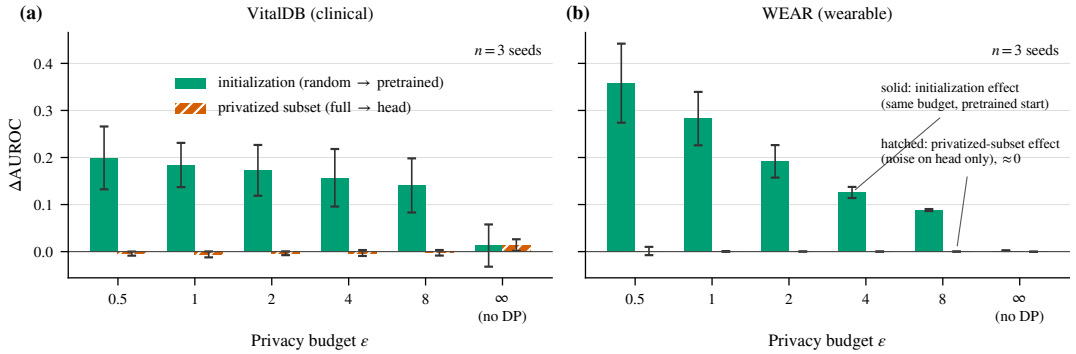


Figure 4: Initialization (B4→B7) vs privatized subset (B7→B6) across budgets.

5.7 Does the hierarchy help? (component ablation)

Question. What does each component (hierarchy, DP, sparsification, quantization) contribute?
What we did. Removed one component at a time; the hierarchy test was pre-registered and run at 6 seeds.
Result. Removing the hierarchy (one flat tier) changes AUROC by -0.0047 $[-0.0088, -0.0006]$, $p = 0.032$ on VitalDB, and -0.0052 $[-0.0074, -0.0030]$, $p = 0.0018$ on WEAR, at matched ε. Sparsification, quantization and loss-aware weighting are well-powered nulls on both corpora.
What it means. The hierarchy does not cost accuracy; removing it costs a small but resolved 0.005. Not claimed as "the hierarchy improves accuracy", because the flat arm also changes the number of local rounds. Compression is free.

5.8 Does multimodal fusion help?

Question. Does combining the two modalities beat the best single modality?
What we did. Fused model vs each single modality, 5 seeds; on eICU also a pre-registered "balance" test that degrades modality B with noise until the two modalities are equally strong.
Result.

- **VitalDB:** fused 0.8653 vs 0.8283, a gain of 0.0370 AUROC (5.9 seed SD) and 0.0671 AUPRC. Resolved.
- **WEAR:** one modality already reaches 0.9976, so there is no headroom to test.
- **eICU, natural:** no gain (+0.0037, within noise).
- **eICU, balanced** (amendment F-A1): fused 0.7282 vs 0.6992, gain +0.0291, positive on 5/5 seeds, paired $p = 0.019$ (control +0.0020).

What it means. Fusion pays when the second modality is competitive; when one modality dominates (eICU's APACHE score), it adds nothing. An early 2-seed figure (+0.0411) shrank at 5 seeds; only the 5-seed figure is quoted.

5.9 The k-law on VitalDB (records per patient)

Question. Does the cost of patient-level privacy grow with records per patient, k?
What we did. Built training sets in which every patient has exactly $k \in \{1, 2, 4, 8\}$ records, and trained a record-level arm and a patient-level arm with the same units per step. VitalDB case-level split, 3 seeds, AUPRC primary.
Result.

k	Gap (record − patient)	95 % half-width	p
1	−0.0071	0.0599	0.661
2	+0.0486	0.0129	0.004
4	+0.0796	0.0163	0.002
8	+0.0872	0.0248	0.004

The curve $A(1 - 1/k)$ with one parameter fits with $A = 0.1015$ (RMSE 0.0119); a two-parameter curve does no better (0.0117). Fitted on $k \leq 4$ only, it predicts 0.0904 at $k = 8$; observed 0.0872. The patient arm is flat (0.508–0.513); the record arm rises (0.502 → 0.596).

What it means. At $k = 1$ a patient is a record, so the two arms must agree; they do. That is a built-in positive control. The growing gap is the record arm exploiting extra records that the patient arm cannot, not the patient arm getting worse.

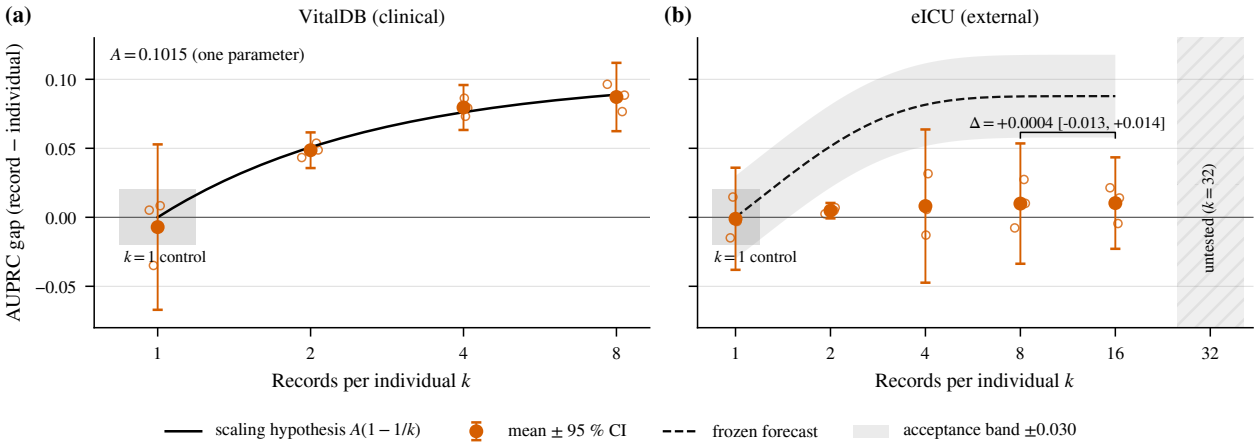


Figure 5: The k-law: (a) VitalDB with the fitted curve; (b) eICU against the frozen forecast.

5.10 External test of the k-law on eICU (pre-registered)

Question. Does the law transfer to a different hospital corpus?

What we did. Before the eICU corpus was built, we froze a forecast from VitalDB, $\text{gap}(k) = 0.0878\,(1 - e^{-0.883(k-1)})$, with an acceptance band of ±0.030, and extended k to 16 to test saturation. ($k = 32$ was dropped by a dated amendment before any result past $k = 8$: only 20 % of eICU patients reach 32 records.)

Result. (after the fix, 3 seeds)

k	Gap	Forecast	p
1	−0.0011 ± 0.0369	0.0000	0.909
2	+0.0048 ± 0.0057	0.0515	0.068
4	+0.0081 ± 0.0554	0.0816	0.593
8	+0.0099 ± 0.0437	0.0876	0.431
16	+0.0103 ± 0.0332	0.0878	0.314

From $k = 8$ to 16 the gap changes by +0.0004 $[-0.0134, +0.0141]$.

What it means. Shape transferred, size did not. The control holds, the gap rises and then flattens, as forecast. But at $k = 8$ it is about one ninth of the forecast, and no gap is significant at three seeds (about 25 seeds would be needed). One hypothesis: eICU's attainable precision is about half of VitalDB's (AUPRC 0.22 vs 0.50 at 4.2 % prevalence). Distinguishing that from a corpus-specific ceiling needs a third corpus. Seeds are *not* being added to chase significance.

5.11 Is the eICU shortfall just the budget? (matched grid)

Question. The forecast was fitted at $\varepsilon = 1$, batch 32; eICU ran at 0.8, batch 64. Is that why eICU is smaller?

What we did. Re-ran the VitalDB k-grid at the eICU operating point.

Result. The VitalDB gaps do not shrink (+0.0914 vs +0.0872 at $k = 8$). But this grid fails its own $k = 1$ control (+0.0398, all seeds the same sign), probably from the smallest pool combined with the largest batch. Correcting every grid for its own $k = 1$ offset attributes about half (51 %) of the $k = 8$ difference to the operating point.
What it means. Under either reading the budget does not explain the factor of nine, and the corpus effect never changes sign.

5.12 Is the cost an artefact of the clipping rule?

Question. Clipping a patient's *summed* gradient hurts patients with many records. Would normalizing by record count before clipping recover the cost? (It changes nothing about the privacy guarantee.)
What we did. Three rules: fixed bound, divide by k, divide by $\sqrt{k}$. Pre-registered criterion: count normalization must recover at least +0.010 AUPRC.
Result.

- **eICU, before the fix:** change -0.0001; the one-sided upper bound +0.0058 excludes +0.010. Fails.
- **VitalDB at $k = 8$, after the fix (D1):** clipped fraction falls from 0.50 to 0.34, AUPRC changes by +0.0038 $[-0.0072, +0.0148]$; the upper bound +0.0113 does *not* exclude +0.010: **inconclusive** by its own rule, though even the bound is one eighth of the 0.0914 cost.
- **eICU, after the fix (D2):** final-round clipped fraction falls from 0.14 to 0.08, AUPRC changes by -0.0012 $[-0.0093, +0.0069]$; the upper bound +0.0043 excludes +0.010. **Fails** (as before the fix). The square-root rule changes AUPRC by -0.0031.

What it means. Normalization changes how often clipping binds but not accuracy. On eICU the clipping rule does not explain the cost; on VitalDB it could explain at most one eighth of it. The cost belongs mainly to the patient unit itself.

5.13 Privacy audit: membership inference (LiRA)

Question. Can an attacker tell whether a *patient* was in training, and does patient-level DP leak less than record-level DP?
What we did. VitalDB person-level split. LiRA with 16 shadow models per arm, 600 member and 600 non-member patients, four attacks (loss threshold, offline LiRA, online LiRA, online LiRA with a global variance), 5 target seeds (3 for the control). A positive control, non-private training of the same head, must show that the attack can detect leakage. Each attack is compared with a "difficulty floor": the AUC from per-patient difficulty alone, without the target model.
Result.

Configuration	Loss AUC	Offline LiRA AUC	Floor
Control (no DP)	0.556	0.605 $[0.528, 0.681]$	0.553
Record-level DP	0.522	0.576 $[0.557, 0.594]$	0.511
Patient-level DP	0.506	0.503 $[0.475, 0.531]$	0.508

The control is detected (online LiRA with global variance 0.597 $[0.586, 0.609]$, entirely above its floor; at three seeds the offline interval spans it). Paired over five seeds, record minus patient: offline LiRA +0.073 $[0.054, 0.092]$ ($p < 0.001$), loss +0.016 $[0.010, 0.023]$ ($p = 0.002$), both surviving correction for four attacks; the two online attacks resolve no difference. True-positive rate at 1 % false positives: 0.024 control, 0.017 both private arms.
What it means. Under offline LiRA, record-level training remains detectably above its floor and patient-level training does not. This is the direction predicted, with a smaller size (0.58, not the predicted 0.7). It is not equivalence with the control (the record arm is below it on two attacks), it rests on two of four attacks, and it is limited to VitalDB. An attack audit bounds what these attacks achieve; it does not prove there is no leakage. On the old case-level split, where members were operations, no difference was resolved.

5.14 Comparison with prior systems (HED-FL, FL2DP)

Question. How does PP-HFM compare with published systems on the same data?

What we did. Re-implemented HED-FL (hierarchical, no privacy, dense updates, 60 % of clients per round) and FL2DP (exponential-mechanism perturbation and shuffling, no hierarchy) inside the same trainer, splits, frozen encoders, rounds and five seeds. FL2DP ran on VitalDB only (about two GPU days per seed on eICU).
Result.

	System	AUROC	Uplink [MB]	Offline LiRA (margin)
VitalDB	PP-HFM ($\varepsilon = 0.9$)	0.8318	252	0.503 (-0.005)
	HED-FL (no DP)	0.8262	560	0.581 ($+0.026$)
	FL2DP	0.8195	1008	0.608 ($+0.049$)
eICU[s]	PP-HFM ($\varepsilon = 0.9$)	0.7936	5235	–
	HED-FL (no DP)	0.8252	12541	–

[s] Source partition, on which the baselines were run (2.9 % of training persons at several hospitals).

On VitalDB PP-HFM exceeds FL2DP by 0.0123 AUROC $[0.0064, 0.0181]$; against HED-FL the AUROC difference (0.0056) is not resolved, AUPRC is +0.030 $[0.005, 0.055]$. In the audit, both baselines have higher attack AUC than PP-HFM under all four attacks. On eICU, HED-FL is 3.17 AUROC points above the private model and level with the non-private reference.
What it means. PP-HFM uploads 2.2–4.0 times less and shows lower measured leakage on VitalDB. On eICU the non-private HED-FL is more accurate; that gap is the price of the patient-level guarantee. FL2DP's candidate set is built from the private gradient, so it satisfies no finite ε, and it also leaks more than HED-FL.

5.15 Private conformal prediction (negative result)

Question. Can each client release a private calibration threshold so that the system outputs prediction sets with 90 % coverage?
What we did. Exponential-mechanism quantile per client ($\varepsilon_{\text{conf}} = 0.1$), aggregated by edges and cloud; coverage measured on the test split.
Result. Thresholds land mid-range (0.42–0.61). The private model's probabilities sit near zero, so every set is just the majority class: coverage equals the base rate (80.1 % on VitalDB, case-level split; 95.8 % on eICU), missing the 90 % target. The non-private threshold does not collapse to the majority class (it covers 36–43 % of positive patients), but its coverage, 80.4–81.8 %, also misses the target.
What it means. With about 100 calibration patients per client at $\varepsilon = 0.1$, the mechanism cannot separate thresholds. Conformal prediction was therefore removed from the system and the contributions, and the full budget $\varepsilon = 0.9$ goes to training. It stays in the paper as an honest negative result.

5.16 Standardization constants: a hidden release

Question. Input normalization means and SDs are fitted on training patients. Do they leak?
What we did. Measured, exactly, how much one patient can move them (leave-one-out over 92,909 eICU individuals), and compared them with constants fitted on the public pool.
Result. Removing one patient moves any mean by at most 0.0017 SD; for 99 % of patients any SD by at most 0.02 %; but one outlier moves one channel's SD by 5.2 %. Pool-fitted constants differ by at most 0.026 SD and 4.4 %.
What it means. Small, but it is a release outside the guarantee. Fix: re-standardize eICU with pool-fitted constants (verified exact: the stored data reproduce the original constants to 4×10^{-8}), re-pretrain the encoders and re-run. That run is the only configuration that is private end to end. It uses a partition in which each of the 2.9 % of training persons treated at two to four hospitals is moved, with all records, to the hospital holding most of their stays, so every person is held by one client and $\varepsilon = 0.9$ holds for all of them (on the source partition they composed to $\varepsilon \leq 1.32$–1.92).
Result. AUROC 0.8208 $\rightarrow$ 0.7820, a cost of 3.88 points $[2.60, 5.17]$ ($p = 0.001$, 5 seeds); AUPRC 2.31 $[1.42, 3.21]$; balanced accuracy 2.80 $[2.52, 3.08]$ (source partition: 3.91 $[2.40, 5.42]$, 2.66, 2.69).
What it means. Public constants make the guarantee end to end and cost about 0.8 AUROC points more than training-fitted ones (3.10). The AUROC and balanced-accuracy intervals lie entirely above the 2-point target (a clear miss); the AUPRC interval spans it.
Status: done 29 Oct.

5.17 Cost of disjoint encoders on VitalDB (F2)

Question. How much does it cost to pretrain VitalDB's encoders on a disjoint pool instead of on client data?
What we did. Stage F2: the patient-level system run on VitalDB (case-level split) repeated with encoders pretrained only on the disjoint pool, five seeds, private and non-private arms.
Result. Against the oracle encoders the private model loses 0.0157 ± 0.0046 AUROC (0.8395 → 0.8237, $p < 0.001$) and 0.0122 ± 0.0262 AUPRC (not resolved). With disjoint encoders the cost of privacy itself is +0.0014 [−0.0033, +0.0062] AUROC and +0.0221 [−0.0003, +0.0446] AUPRC. (Before the fix, record-level: 0.0322 AUROC, 0.0763 AUPRC.)
What it means. Honest encoders cost about 1.6 AUROC points on VitalDB, half the earlier estimate. Standardization constants are still fitted on client data here, so this run is not end to end; only the eICU re-run with public constants is.

6 Mistakes found and fixed

A large part of this project was finding errors that produced plausible-but-wrong numbers without any crash. Each one was fixed, tested, and the affected results re-run. The important ones:

Error	Effect and fix
Noise calibrated from one client	Every DP arm exceeded its budget. Now calibrated against all clients.
Conformal calibration fired 5×	Total ε was 3.49, not 0.96. Fixed to fire once; later removed entirely.
NS-3 had never run	Every early "network result" was an analytical fallback. NS-3 built; packet-level every round.
Stale partition metadata	Noise calibrated on wrong counts. Rebuilt.
eICU training collapse	Model decayed to constant output in long epochs, which would have poisoned every eICU k-number. Fixed validation cadence.
Divided by the realized batch $\|S\|$ (22 Oct)	Not the analysed mechanism; at small hospitals it reveals the patient count. Now divides by the expected batch. All patient-level results re-run in stages A–G.
Headline runs were record-level	The system runs used record-level DP while claiming patient-level. Re-run at patient level.
Aggregation weights leaked sizes	Now static declared sizes.
VitalDB split by surgical case (30 Oct)	55 of 940 test persons also had a training case, and 68 training persons were on two clients. Rebuilt the split by person; privacy cost, unit cost, baselines and audit re-run (Section 4).
Objective scores out of date (1 Nov)	The scoring files used an old latency benchmark and scored utility on accuracy. Re-scored from the final runs; the old files are kept.

Why this matters. Every one of these would have survived review unnoticed and would have broken the central claim. The paper's numbers are all from the corrected code; the old results are archived, not overwritten.

7 Honest scorecard: what is and is not claimed

Claim	Status
Patient-level guarantee for hierarchical multimodal FL, all tiers accounted	proven for the training transcript (Theorem 1); end to end only for eICU with pool-fitted constants, on the partition with one hospital per person
Communication, latency, energy objectives	met on all three corpora (latency: workstation CPU; energy: modeled; network: simulated)
No utility cost on VitalDB at $\varepsilon = 0.9$	supported, person-level split (-1.18 $[-1.63, -0.73]$)
Utility cost ≤ 2 points on eICU	inconclusive (3.10 $[1.55, 4.65]$); **not met** end to end (3.88 $[2.60, 5.17]$)
eICU update delivery $> 99\,\%$	**fails** at 2 edges (97.83 %); met from 8 edges
Pretrained initialization makes strict privacy survivable	strongly supported (VitalDB case-level, WEAR)
Head-only DP improves accuracy	**refuted** (it helps communication)
k-law shape on VitalDB	supported, with positive control (case-level split)
k-law transfers to eICU	shape yes, magnitude no, unresolved
Cost is not a clipping artefact	eICU yes; VitalDB inconclusive ($\leq 1/8$ of cost)
Patient DP leaks less than record DP (MIA)	supported on VitalDB under offline LiRA and loss (2 of 4 attacks); smaller than predicted
Less traffic and lower measured leakage than HED-FL, FL2DP	supported on VitalDB; on eICU non-private HED-FL is more accurate
Private conformal prediction	**failed**, removed
Fusion helps	VitalDB yes; eICU only when balanced
Hierarchy costs accuracy	no (removing it costs 0.005)

None of the building blocks is new (user-level DP, private fine-tuning, hierarchical FL). The contribution is their composition inside a hierarchy with a proof, and careful *measurement* of what it costs.

8 Still to do

No experiments are pending. What remains is approvals and submission work.

Item	Status
Author block	college grid now; IEEE IoT Journal needs the journal layout (one switch in `main.tex`); author order and ORCID iDs for all authors
Page length	19 pages; IoT Journal charges for pages beyond eight
AI-use disclosure	to be written in the acknowledgment
Code and results release	staging copy built and validated; waits for the WEAR licence check, Apache-2.0 approval by all authors and MAHE, and IEEE rights
Optional	eICU membership-inference audit (about 160 GPU hours); more seeds only as a separately declared replication

9 Glossary

Federated learning (FL) Training where data stays at each site and only model updates are shared.
Hierarchical FL FL with an intermediate edge tier that aggregates nearby clients before the cloud.
Differential privacy (ε, δ) A guarantee that the output barely changes whether or not one unit's data is included. Smaller ε is stronger; $\varepsilon \leq 1$ is considered strong.
Privacy unit What the guarantee protects: one record, or one patient.
DP-SGD Training with per-unit gradient clipping and Gaussian noise.
Clipping norm C The maximum size of one unit's contribution.
Noise multiplier σ Noise scale relative to C.
Accountant (PRV, RDP) Computes the total ε over all steps.

AUROC / AUPRC Ranking quality of the model; AUPRC is more sensitive when positives are rare (eICU: 4.2 %).

Seed One independent repetition of a training run; paired seeds compare two arms on identical splits.

Pre-registration Writing down the prediction and the pass/fail rule before running the experiment, so the result cannot be reinterpreted later.

LiRA Likelihood-ratio membership-inference attack using shadow models.

Oracle vs disjoint encoders Pretrained on client data (upper bound, not private) vs on a separate public pool. Disjoint encoders are needed for the guarantee to hold end to end, together with public standardization constants.

Inconclusive The confidence interval contains the target, so the data cannot decide pass or fail at the fixed number of seeds. It is neither a pass nor a failure.